W3C2 WEP 30/10/24

Slide 53 - 59

Paket lass

Throughput Battleneck

Slide 60 -

Security

sniffing

spoofing

Denial of service (DoS)

Slide 65 imp

10 cars 12secs

120 secs

Speed = 1000

distance = 100km

10 ~~secs~~ mins

A $\longrightarrow$ b

$\frac{100}{1000}$ 0.1h

6 min

W4C2 Unlimited protocal

Slide 67

Application → build connection first, Reliant conn...

Transport → TCP/UDP → unreliable video streaming

Network → 1 protocol IP

source se destination tak pohchana

32 bit / 128 bit

IPV4 IPV6

X

Data link → MAC

48 bit

use to find next device

physical → Converting into bits

3.0 3-61

packet send krne ke baad timer chala ta nai taka reasanable amaunt wait karnai ke baad pkt dobara send kren.

3-64

3-65 Exam

rdt 3.0

Perfamnce X

ek class miss hui

19/Oct/24

Chapter 2 slide 18

Stateless & stateful

Persistant & non persistent

2-44 formula

4 methods

Status code (not much dicussed)

2-32 Cookies

35 USES

HTTP is statless so they use Cookies

Challenges

Traking

2-42 web cache

Cache example

no. ~~of~~ calculation

E-mail

SMTP Pus

POP3/IMAP Pu

Domain nam

Router any un

~~A~~ implemented

network can

edges ko

DNS

~~IP~~ host

why

tiplexing

Exam 5marks (scenario based) 26|Nov|

Network devices (diagram)

Organisation of Network

- End user devices & Network Devices
- connectivity Devices
- Five categories of Network Devices
- Repeater * (discussed in detail)
- Function of repeater

Attenuation : signal weak

Token (jis ke pas token hoga wahi data send)

* HUb

Types of hub

* what is a bridge

> switch

> 2 Layer switch

> 3 layer switch

> Routers

> Routing Principle

> Difference

> Gateway

→ build connection
first
Reliable connection
TCP/UDP
↓
unreliable
video streaming

find next device

ls

7/Nov/24

Chapter 2 Slides

, Applications are for end systems only not for network core

, **Client server paradigm**

client ⇄ server ⇄ client

• **Peer - Peer architecture (P2P)**

no server included

clients ⇄ clients
$\boxed{\text{files}}$

Seeds **Leaches**

Eg Bittorrent U torrent

• **Processes communicating**

programing running within a host

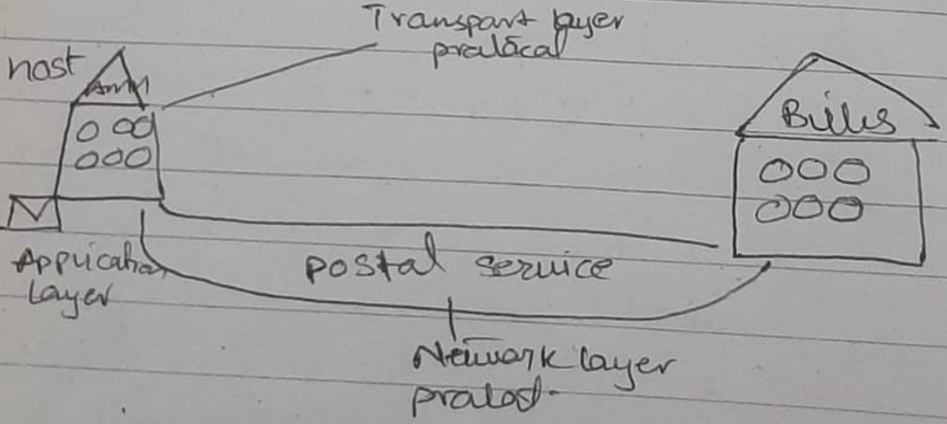
example student communicating each other in a class using inter process communication

• **Client process** initiat a request

2-71
Local DNS name server (fridge)

2-72
iterated query (Exam)
2-73
recursive query (Exam) diagram
2-74
Caching DNS information
TTL (Expire) (Time to Live)

3-5 (Exam style)



Tranport layer protocal
3-9
Two principle internet transport protocal
TCP & UDP

3-11
multiplexing |

3-13
port wi

IP

3-25

Connection-oriented demultiplexing

3-27

(fridge)

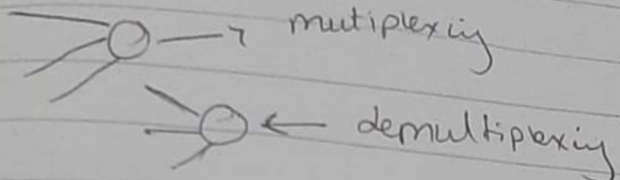
diagram

3-11

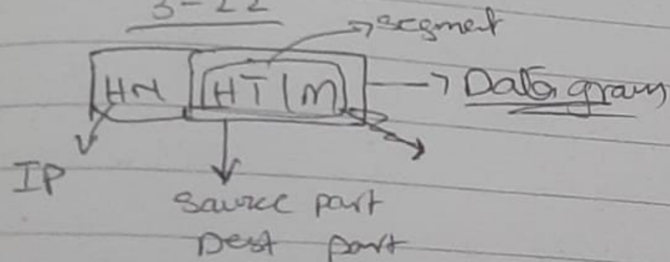
multiplexing / demultiplexing → directing

<u>3-13</u>

port will recognise kaha jaiga



<u>3-22</u>



E-mail

SMTP push Simple mail transfer

POP/IMAP push

Domain name system 2-64 DNS

Router only understand IP

~~A~~ implemented on Application layer

network core change nahi kiya

edges ko change kiya

DNS 2-65

~~IP~~ host name to IP Address

why DNS is distributed ? Exam

2-66

not much discussed

2-67

Levels discussed

2-69

not much discussed

2-70

Top lvl domain

authoritative

4/Dec/24

Reliable data transfer

3-44, 3-45, 3-46, 3-47, 3-48, 3-50, 3-52

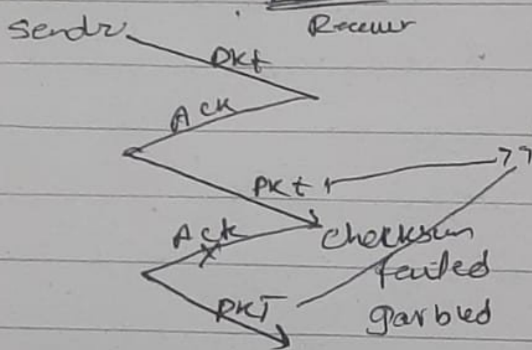
3-~~45~~

Exam

Q How you will develop reliable data transmission

3-55 3-56 (Draw)

FSM



same packet

Tail sequence use kiya nai

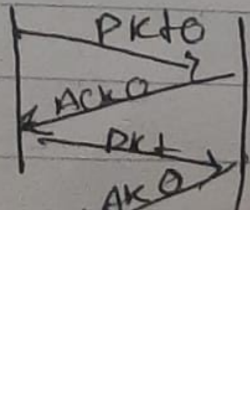
2.1 3-58

Q why sequence os use 0 & 1

Ans • we dont need

• Discard duplicate

3-59 2.2 NAK



3.0

packet sen

nai taka

ke bad

3-64

3-65

rdt 3

Perfor